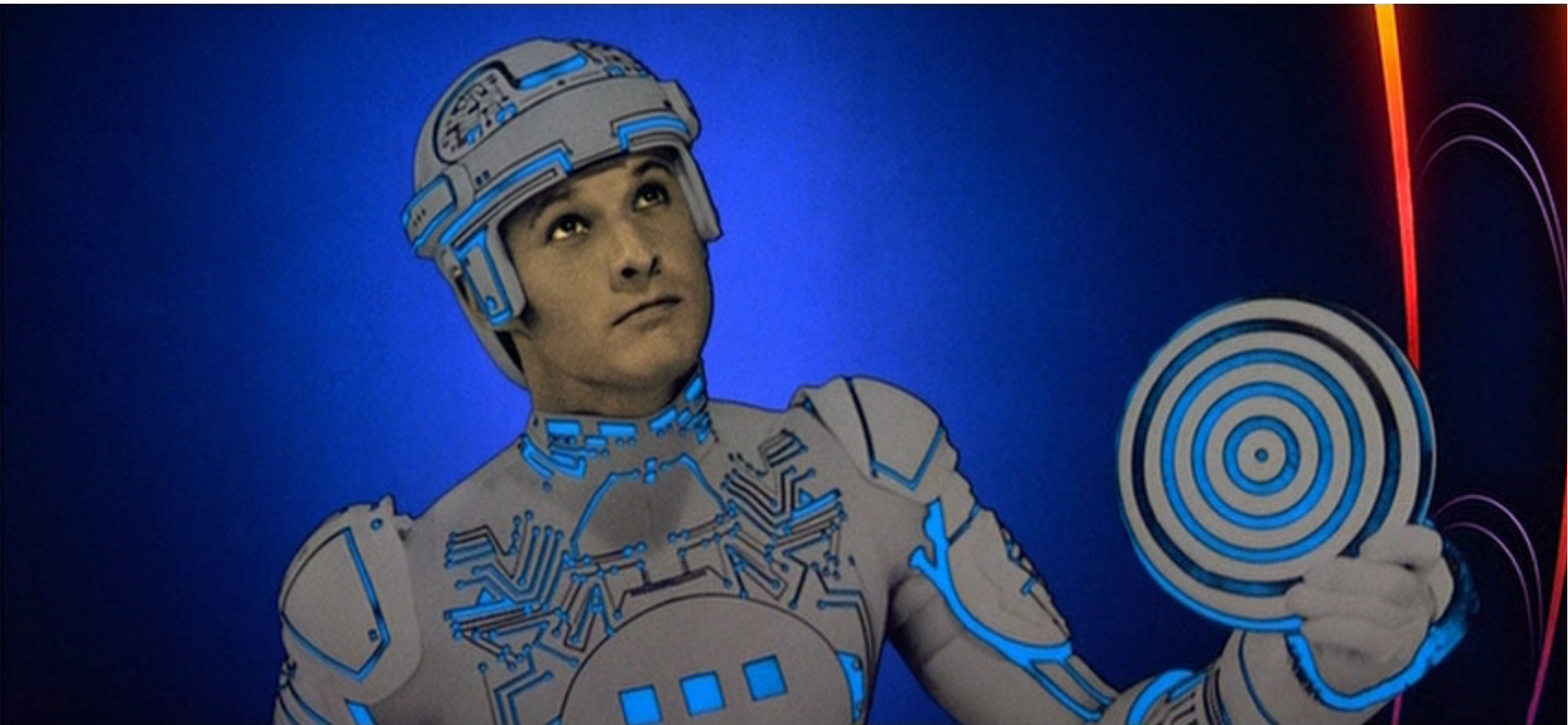


Implications of Functional Programming on the Human Rights

christo@cryptodrunks.net



Kevin Flynn: Who's that guy?

Warrior #1: That's Tron.

Transformation through collaboration,
ugly and otherwise, is the human condition.

– *Anna Lowenhaupt Tsing*

Organizations which design systems ...
are constrained to produce designs
which are copies of the communication
structures of these organizations.

– *Melvin Conway*

Syrian Archive

Curating Visual Documentation for Justice and Human Rights

The Syrian Archive is a Syrian-led and initiated collective of human rights activists dedicated to curating visual documentation relating to human rights violations and other crimes committed by all sides during the conflict in Syria with the goal of creating an evidence-based tool for reporting, advocacy and accountability purposes.

<https://syrianarchive.org>

Warning: this video may contain graphic content



The video shows footage before and during the attack on the hospital, the transfer of patients and the damage at the targeted site

[Download file ↓](#)

Online Link

<https://www.youtube.com/watch?v=1xoI1ZXbUc4>

Meta

md5 b03cb96c7d8963e5a34efd471ef02643 - acquired
2018-02-06T12:08:24.721Z

Incident Occurred at:

2018-02-05

10:19:56

Location:

IDLIB : Kafranbel nahiyah

Precise Location:

Acquired From:

Mohamed Aldaher

Weapons Used:

Collections:

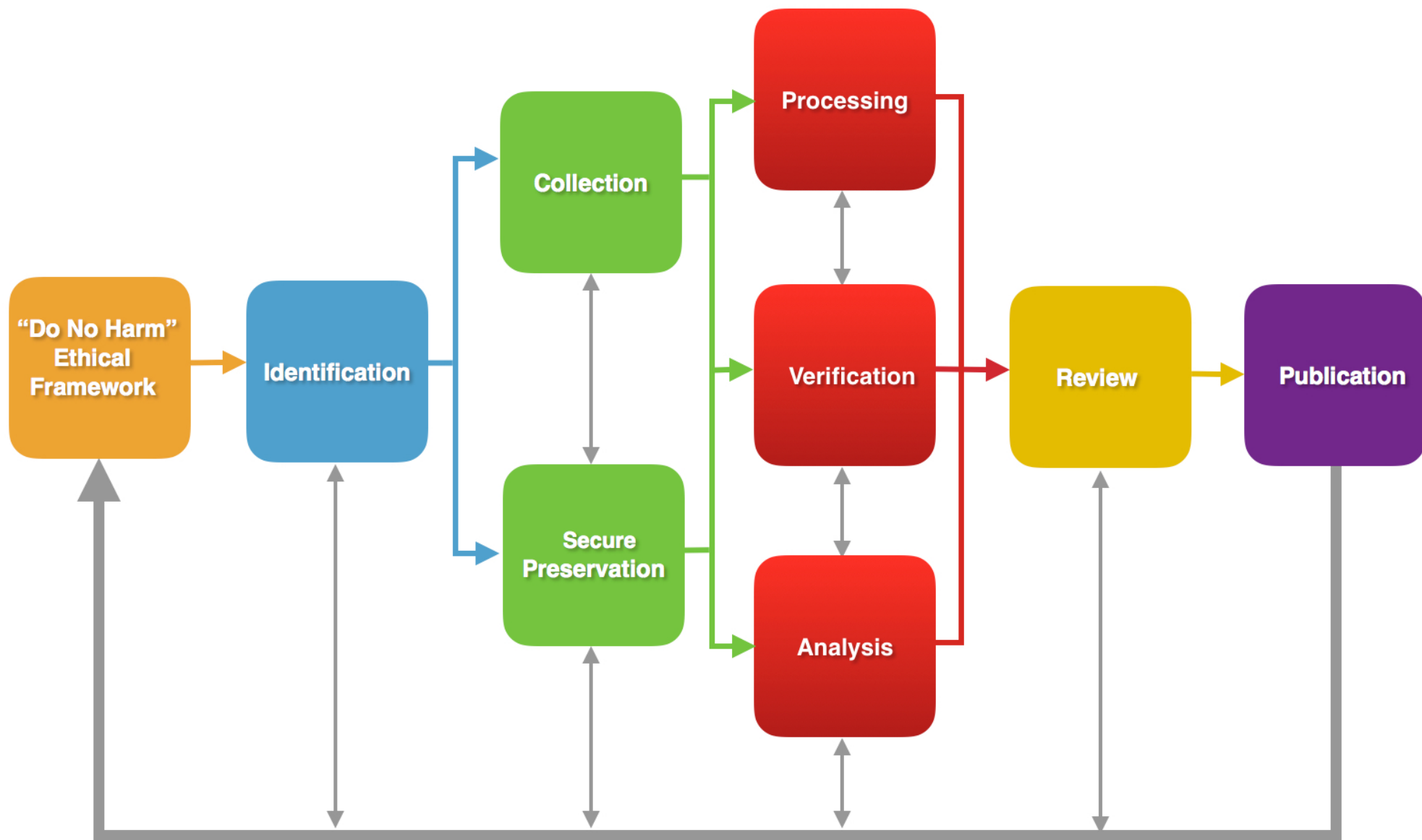
Attacks against hospitals

Type of Violation:

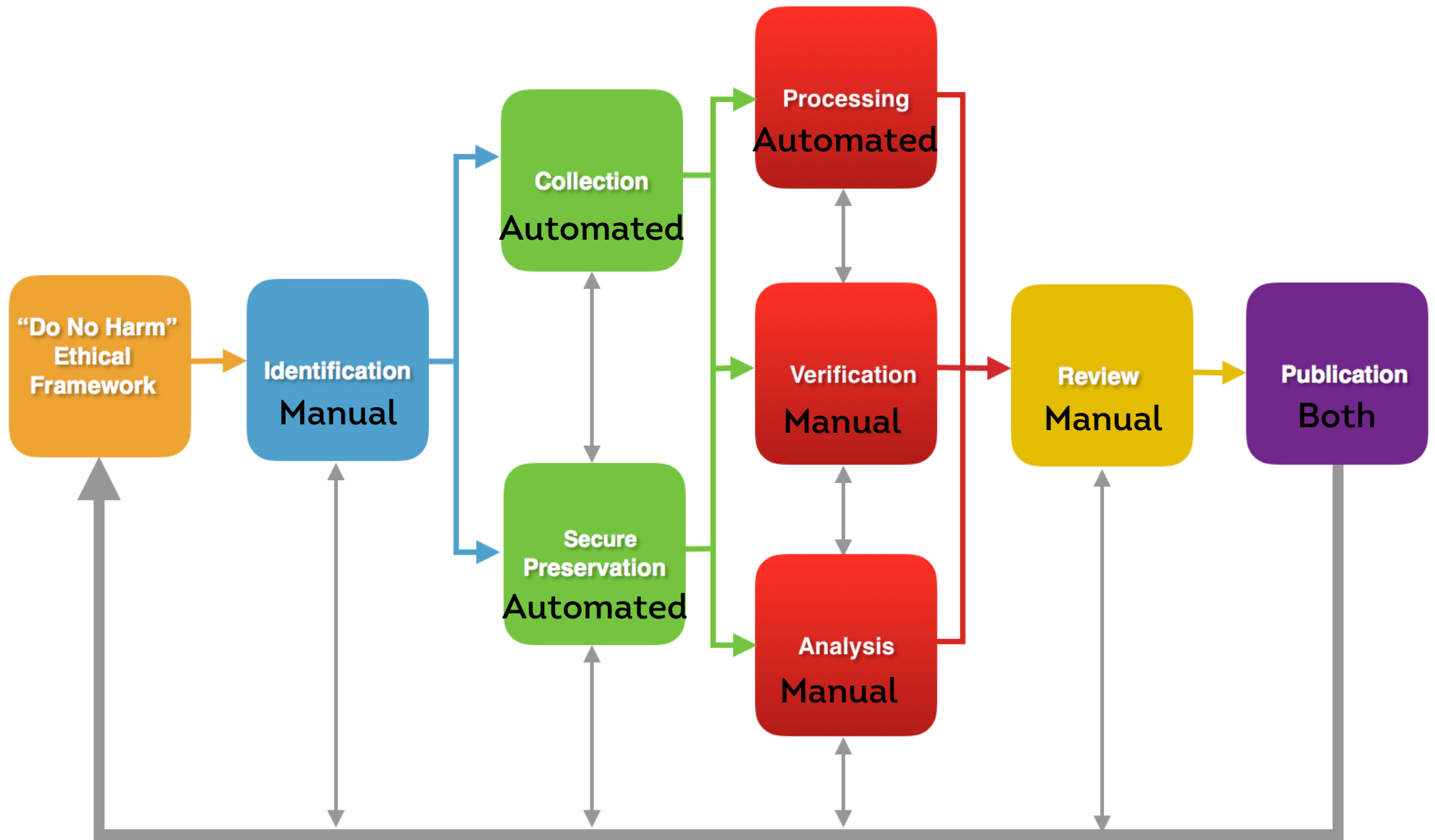
Violations against specifically protected persons and objects

<https://syrianarchive.org>

Syrian Archive Digital Evidence Workflow



Syrian Archive Digital Evidence Workflow



- Allow to rearrange it's elements.
- Play nice in a bigger context.
- Embed in a workflow.
- Preserve peoples agency.

**Theory is always a detour on the
way to something more important.**

$$(g \circ f)(x) = g(f(x))$$

**AN INTRODUCTION
TO THE
THEORY OF LISTS**

by

Richard S. Bird

Oxford University
Computing Laboratory
Programming Research Group-Library
8-11 Keble Road
Oxford OX1 3QD
Oxford (0865) 54141

Technical Monograph PRG-56

October 1986

**Oxford University Computing Laboratory
Programming Research Group
8-11 Keble Road
Oxford OX1 3QD
England**

SugarCube - Monoidal data sequences.

<https://gitlab.com/sugarcube/sugarcube>

$f \rightarrow \text{twitter_feed}$

$g \rightarrow \text{tika_location}$

$h \rightarrow \text{exif_extract}$

$k \rightarrow \text{csv_expand}$

Sequence $\rightarrow k(h(g(f(x))))$

- Sequences are lists of functions.
- Data is a list of records.

```
2018-02-23T06:19:13.240Z - info: Starting run 4995368760c98d9981b120aa48a68c7e4dd27df9.
2018-02-23T06:19:13.248Z - info: Starting the sheets_queries plugin.
2018-02-23T06:19:13.250Z - info: Fetching 1 sheet
2018-02-23T06:19:15.217Z - info: Expanded 108kFL7R46ErCo45qABcTCU6rrhWsXVCJWYWyENTEz7E/Sec Queries to 4 queries.
2018-02-23T06:19:15.218Z - info: Fetched 4 queries.
2018-02-23T06:19:15.222Z - info: Finished the sheets_queries plugin.
2018-02-23T06:19:15.223Z - info: Starting the sec_search plugin.
2018-02-23T06:19:15.225Z - info: Searching the SEC for 'Cobalt Refinery'
2018-02-23T06:19:22.768Z - info: Fetched 29 results.
2018-02-23T06:19:22.939Z - info: Finished the sec_search plugin.
2018-02-23T06:19:22.939Z - info: Starting the tika_location plugin.
2018-02-23T06:19:24.878Z - info: Finished the tika_location plugin.
2018-02-23T06:19:24.879Z - info: Starting the regex_match plugin.
2018-02-23T06:19:24.880Z - info: Matching: /((\w+\s*)?(\w+\s*)?Refinery(\s*\w+)?(\s*\w+)?)|((\w+\s*)?(\w+\s*)?Capacity(\s*\w+)?(\s*\w+)?)|((\w+\s*)?(\w+\s*)?T
ons(\s*\w+)?(\s*\w+)?)/gi
2018-02-23T06:19:25.358Z - info: Finished the regex_match plugin.
2018-02-23T06:19:25.359Z - info: Starting the sheets_export plugin.
2018-02-23T06:19:26.868Z - info: Units exported to https://docs.google.com/spreadsheets/d/108kFL7R46ErCo45qABcTCU6rrhWsXVCJWYWyENTEz7E/edit#gid=0
2018-02-23T06:19:28.399Z - info: Merging 13 new units and 536 existing rows.
2018-02-23T06:19:32.423Z - info: Finished the sheets_export plugin.
2018-02-23T06:19:32.425Z - info: receiving stats for: none
2018-02-23T06:19:32.427Z - info: Finished the LSD.
```

*\$ sugarcube -q queries.json -c config.json *
-p twitter_feed,tika_location,exif_extract,csv_export

- Not Turing complete, nor off the shelf, somewhere in between.
- Compose functions to form chains of executions.
- Lists as the fundamental data structure.
- Rearrange compositions depending on needs and insights.
- Reproduceable processes and workflows.

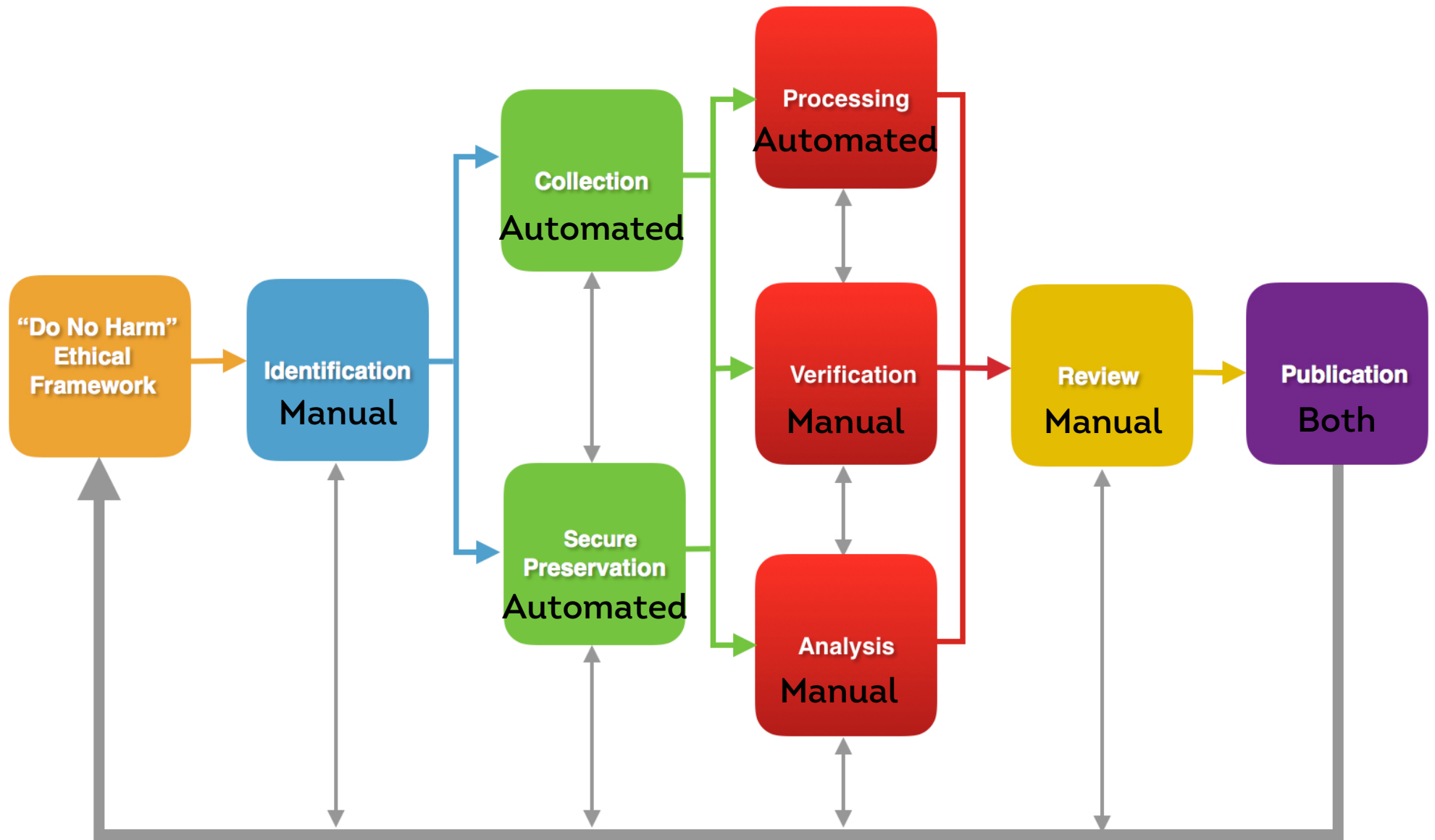
Traps

- Scalability is the ability to expand a project without having to change its framing assumptions.
- Scalability requires that project elements are oblivious to the indeterminacies of encounters.
- Scalability prevents meaningful diversity, that is, diversity that might change things.

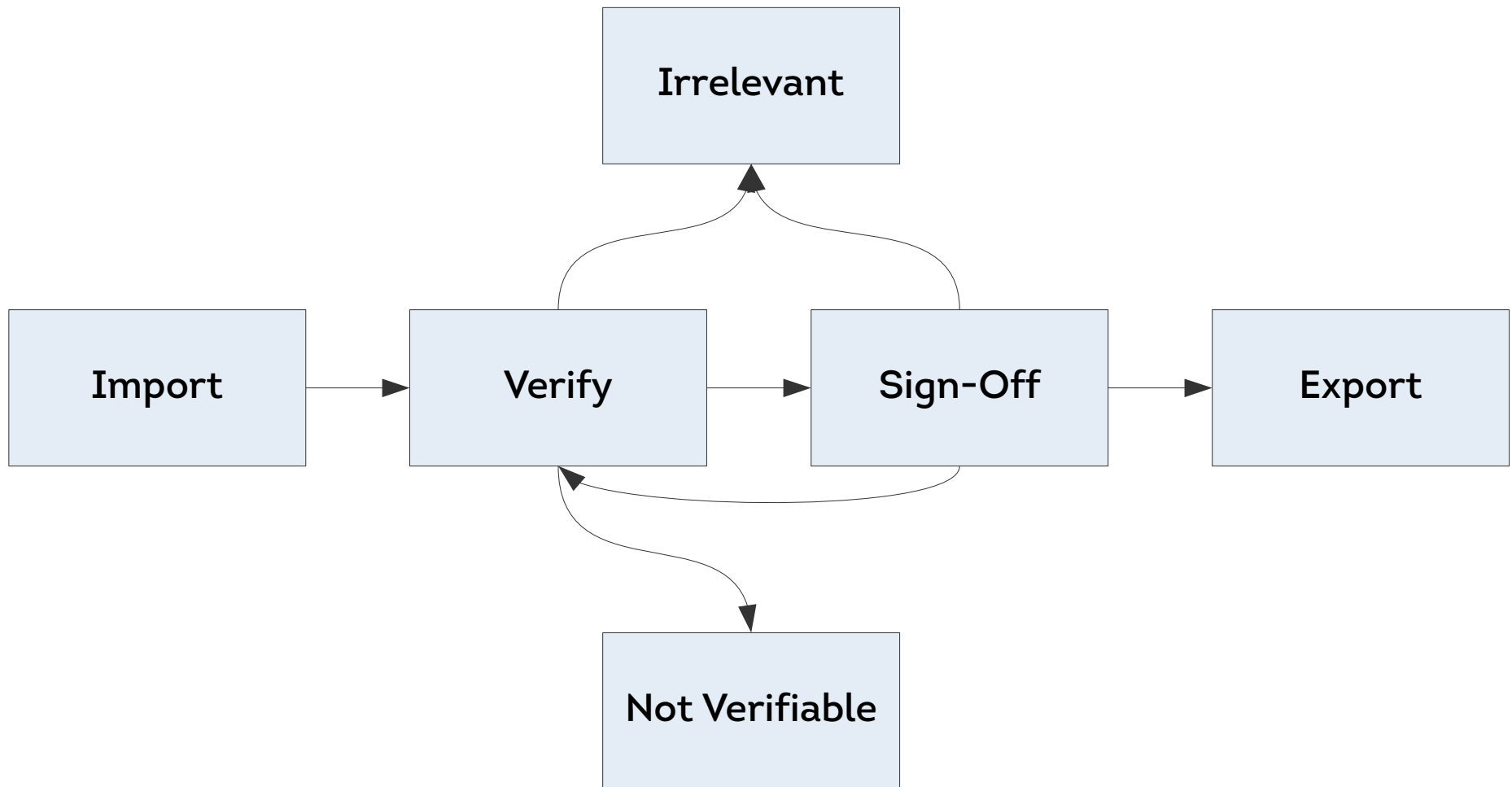
Traps

Techno-fixes

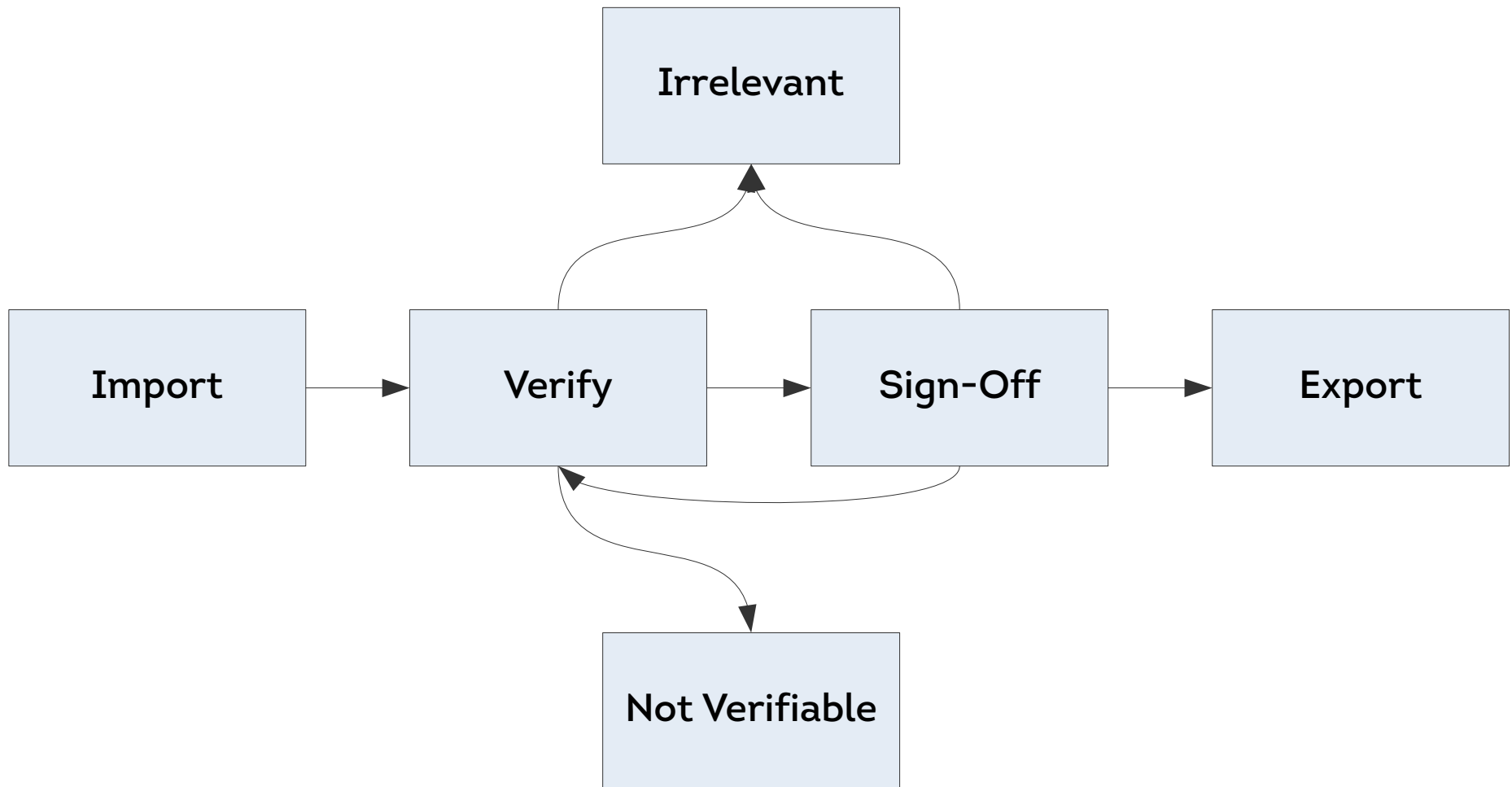
Syrian Archive Digital Evidence Workflow



Steps of Verification



Steps of Verification



State transitions form a data sequence.

Conclusion

- Lift secondary effects to become first class.
- Abstractions in your code are affecting users as well.
- Composition and decomposition give a vocabulary for the ubiquitous language of your domain.